Linux Servers Monitoring using Grafana, Prometheus, and Node Exporter

Prometheus

Create Prometheus User and Directories

- Create a system user for Prometheus without login access and sets up configuration and data directories.

```
useradd --no-create-home --shell /bin/false prometheus
mkdir -p /etc/prometheus /var/lib/prometheus
```

Set Permissions

- Gives ownership of Prometheus directories to the Prometheus user for proper access control.

```
chown -R prometheus:prometheus /etc/prometheus
/var/lib/prometheus
```

Download and Extract Prometheus

- Downloads the latest Prometheus binary and extracts it for installation.

```
wget
https://github.com/prometheus/prometheus/releases/download/
v3.7.3/prometheus-3.7.3.linux-amd64.tar.gz
tar -xvf prometheus-3.7.3.linux-amd64.tar.gz
```

Move Prometheus Binaries

- Moves Prometheus executables to the system path for global access and assigns ownership.

```
mv prometheus promtool /usr/local/bin
chown prometheus:prometheus /usr/local/bin/prometheus
/usr/local/bin/promtool
```

Move Configuration Files

- Moves configuration and console files to the Prometheus config directory with proper permissions.

```
mv prometheus.yml /etc/prometheus/
mv consoles console_libraries /etc/prometheus/
chown -R prometheus:prometheus /etc/prometheus/
```

Configure Prometheus

Edit the configuration file:

```
vim /etc/prometheus/prometheus.yml
```

Sample Configuration:

```
global:
  scrape_interval: 10s

scrape_configs:
  - job_name: 'prometheus_master'
    scrape_interval: 5s
    static_configs:
      - targets: ['localhost:9090']
```

Create Systemd Service File

- Creates a service to automatically start Prometheus on system boot.

```
vim /etc/systemd/system/prometheus.service
```

Service Configuration:

```
[Unit]
Description=Prometheus Monitoring System
Wants=network-online.target
After=network-online.target

[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/var/lib/prometheus/ \
  --web.console.templates=/etc/prometheus/consoles \
  --web.console.libraries=/etc/prometheus/console_libraries

[Install]
WantedBy=multi-user.target
```

Enable and Start Prometheus

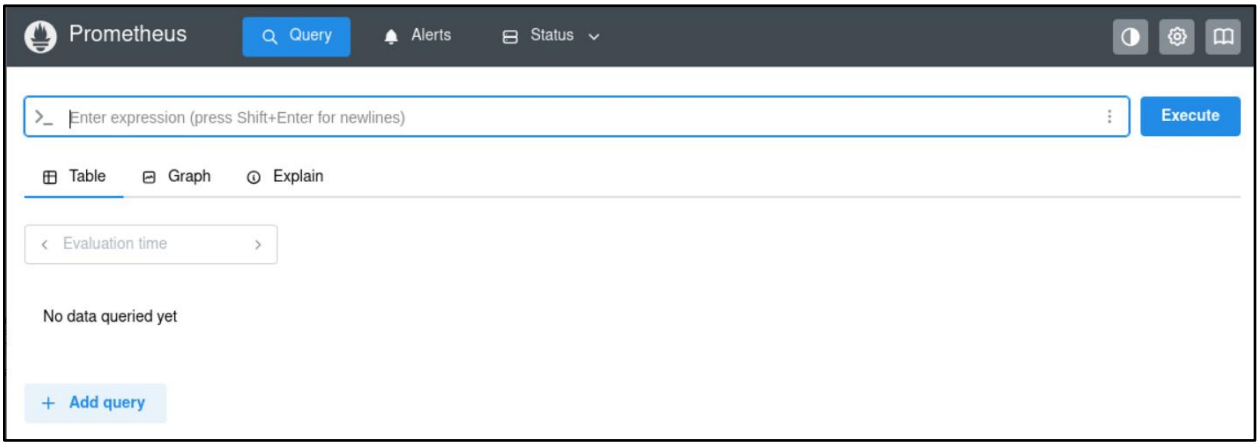
- Reloads systemd, enables Prometheus service, and verifies its running status.

```
systemctl daemon-reload
systemctl enable --now prometheus
systemctl status prometheus
```

Access Prometheus Web Interface

- Open Prometheus dashboard in a browser to verify installation.

```
http://<server_IP>:9090
```



Install Node Exporter

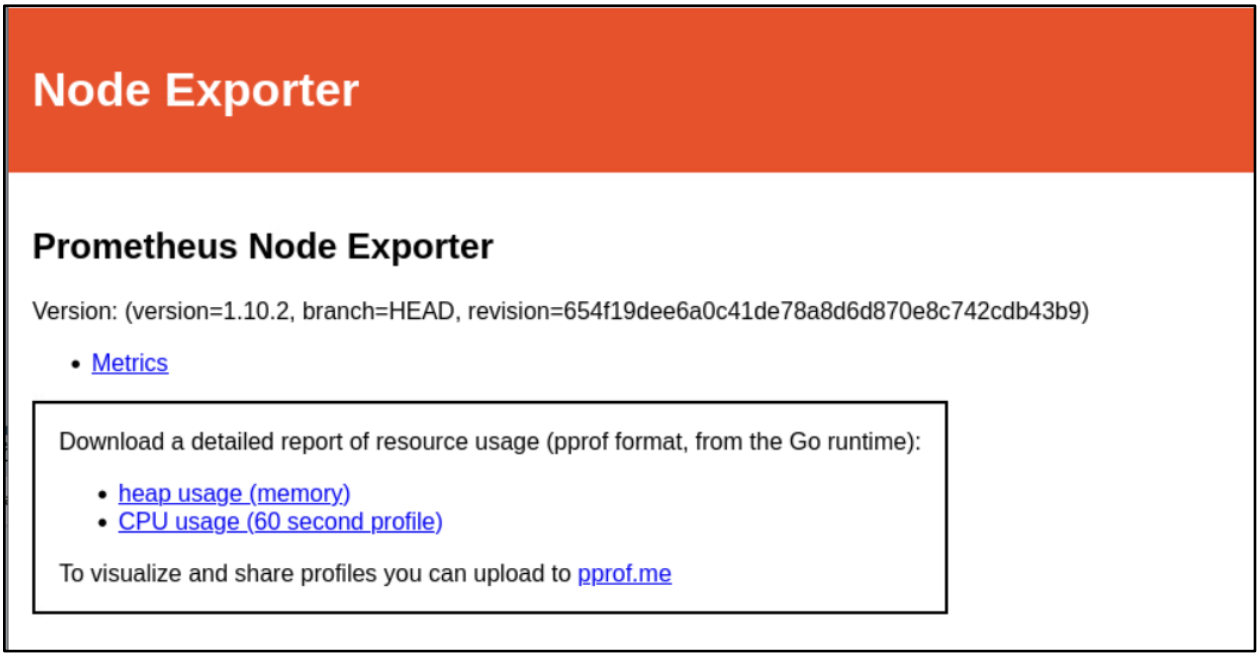
- Downloads and runs Node Exporter to collect system metrics such as CPU, memory, and disk usage.

```
wget
https://github.com/prometheus/node_exporter/releases/downlo
ad/v1.10.2/node_exporter-1.10.2.linux-amd64.tar.gz
tar -xvf node_exporter-1.10.2.linux-amd64.tar.gz
cd node_exporter-1.10.2.linux-amd64
./node_exporter
```

Verify Node Exporter

- View raw system metrics exposed by Node Exporter.

```
http://localhost:9100/metrics
```



Add Node Exporter to Prometheus Configuration

Edit Prometheus config:

```
vim /etc/prometheus/prometheus.yml
```

Add the Node Exporter target:

```
scrape_configs:
  - job_name: 'node'
    static_configs:
      - targets: ['localhost:9100']
```

Restart Prometheus

- Applies the new configuration and starts collecting node metrics.

```
systemctl restart prometheus
```

Verify Targets in Prometheus

- Check that both Prometheus and Node Exporter targets are listed and active.

```
http://localhost:9090/targets
```

Grafana

Install Grafana (Dashboard Visualization Tool)

- Imports Grafana's GPG key for package verification.

```
wget -q -O gpg.key https://rpm.grafana.com/gpg.key
sudo rpm --import gpg.key
```

Configure Grafana Repository

- Adds Grafana repository for installation via DNF.

```
vim /etc/yum.repos.d/grafana.repo
```

Repo File:

```
[grafana]
name=Grafana
baseurl=https://rpm.grafana.com
repo_gpgcheck=1
enabled=1
gpgcheck=1
gpgkey=https://rpm.grafana.com/gpg.key
sslverify=1
sslcacert=/etc/pki/tls/certs/ca-bundle.crt
```

Install and Start Grafana

- Installs Grafana and enables it to start automatically on boot.

```
sudo dnf install grafana -y
sudo systemctl enable --now grafana-server
```

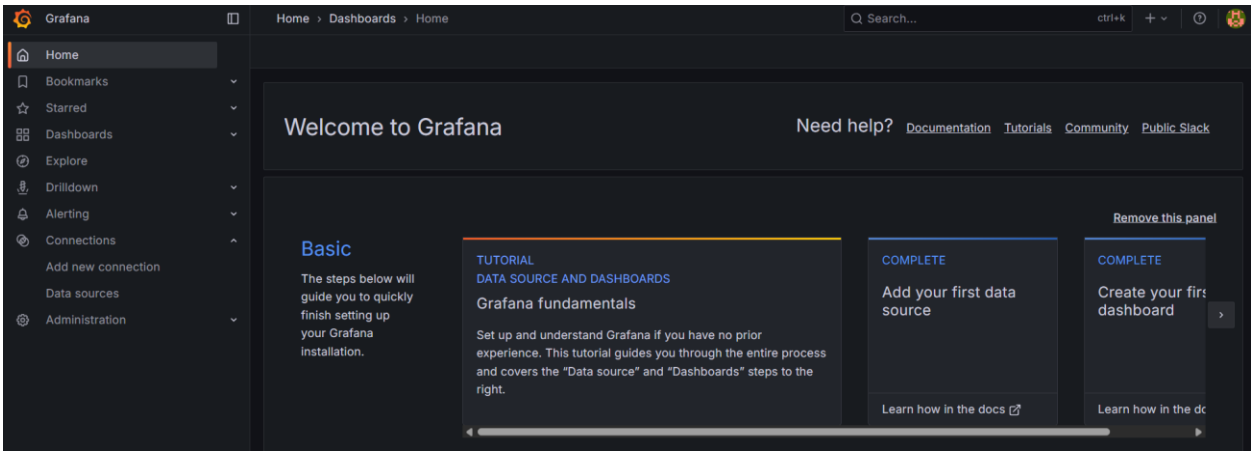
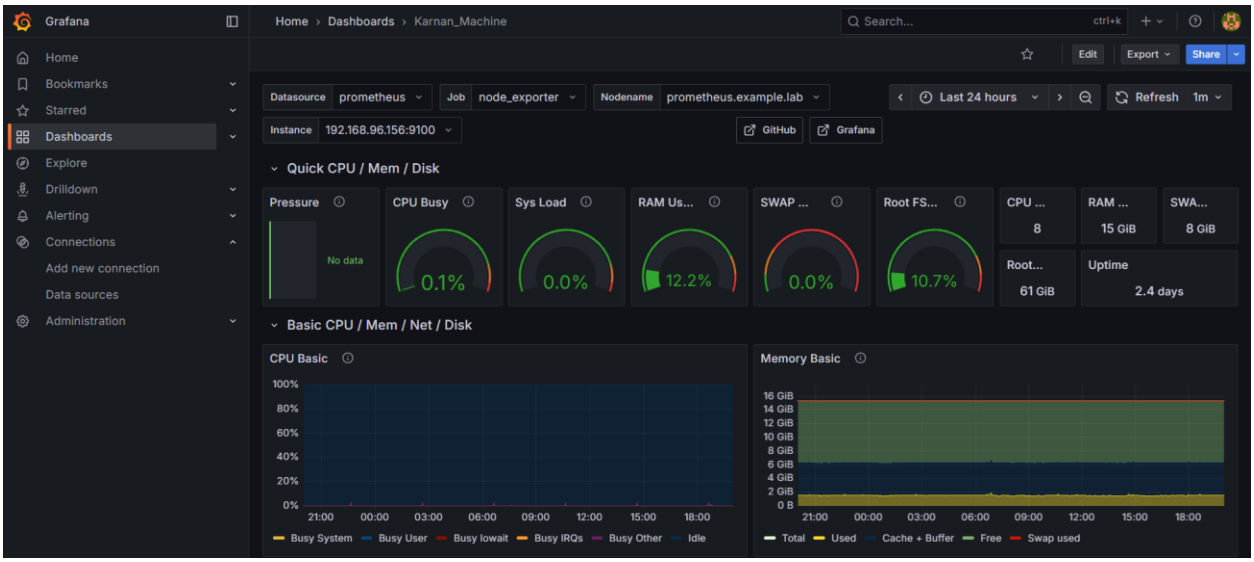
Access Grafana Web Interface

- Log in to Grafana and connect Prometheus as a data source to create dashboards.

```
http://<server_IP>:3000
```

Default Credentials:

- Username: admin
- Password: admin



- This is a Grafana dashboard connected to Prometheus and Node Exporter, showing system performance metrics for your monitored machine (prometheus.example.lab).